

ECOLOGY

Introduction to Basic Ecology

Ecology—the scientific study of interactions between different organisms and between organisms and their environment or surroundings.

- **Ecology** (from Greek: οἶκος, "house" or "living relations"; -λογία, "study of") is the scientific study of the distributions, abundance and relations of organisms and their interactions with the environment. Ecology includes the study of plant and animal populations, plant and animal communities and ecosystems.
- Ecology is a branch of biology concerned with understanding how organisms relate with each other and their environment. This branch of biology mainly deals with the relationships between the organisms, their relationships among each other, their relationships towards the shared resources, their relationships with the space they share, and even their relationships with the non-living aspects in the environment.

- **Ecosystems** describe the web or network of relations among organisms at different scales of organization.
- Ecologists study organisms and their interactions with the environment, from bacteria involved in nutrient recycling to tropical rainforests that influence Earth's atmosphere.
- The discipline of ecology emerged from the natural sciences in the late 19th century.
- Ecology is not synonymous with environment, environmentalism, or environmental science. Ecology is closely related to the disciplines of physiology, evolution, genetics and behavior.

ECOSYSTEM

British ecologist **Arthur Tansley** first defined the term Ecosystem.

Definitions:

The biotic community and the abiotic environment together constitute the ECOSYSTEM.

Ecosystem is a dynamic system where the biotic and abiotic components are constantly acting and reacting upon each other bringing forth structural and functional changes.

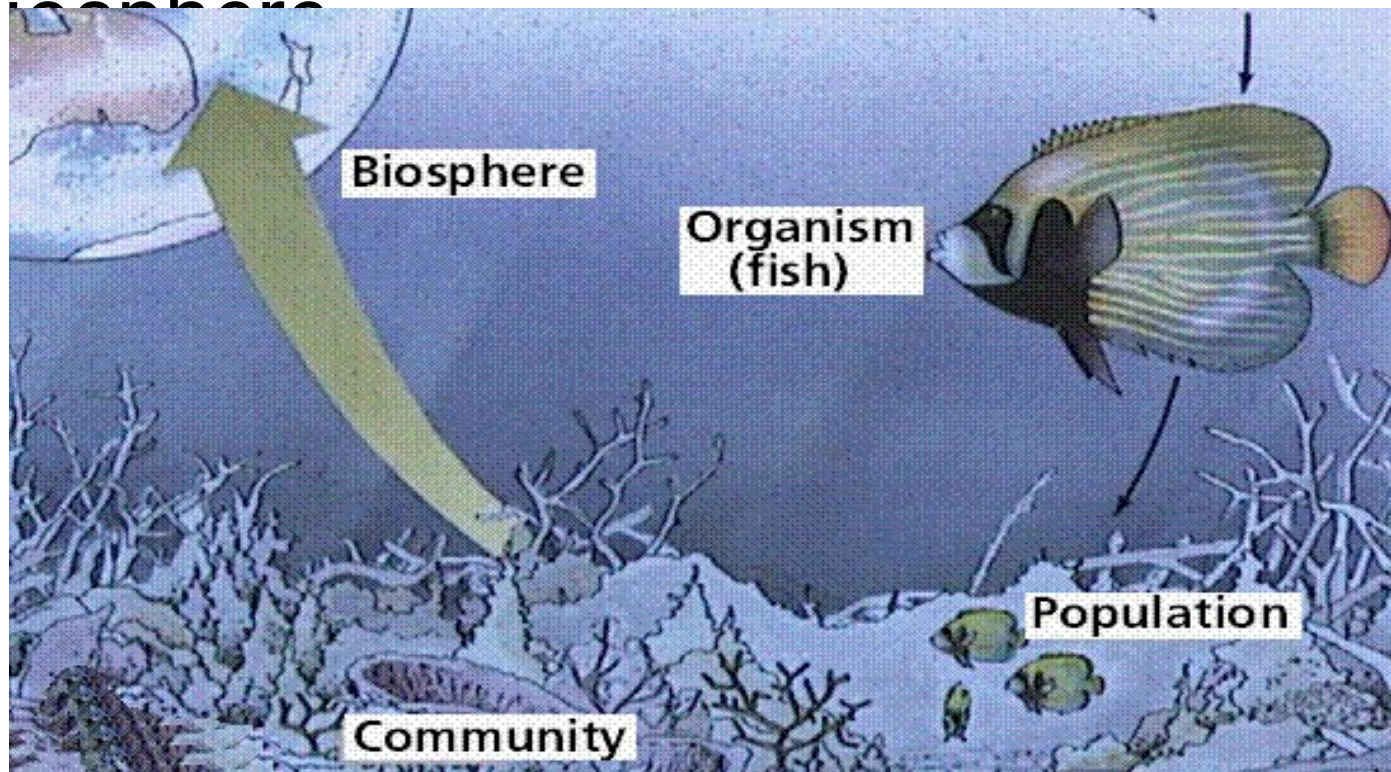
Ecosystem is a complex interacting system of communities and their physical environment.

Importance of Ecology

- Ecology is essential in providing the information on how the world works. It also provides facts on the interdependence between the natural world and people.
- A better understanding of ecological systems can help us understand the global and regional consequences of competition among humans for the scarce natural resources that support us. Ecology-
 - ✓ helps in environmental conservation
 - ✓ Ensures proper resource allocation
 - ✓ Enhances energy conservation
 - ✓ Promotes eco-friendliness
 - ✓ Aids in disease and pest control

Ecological Organization

- Ecological relationships range from an individual organism to the entire biosphere



Ecological Organization

Organism- A single living thing

Population-A group of organisms of the same species that live together.

Community-A group of Populations that live together

Ecosystem- Communities and their physical environments

Biome- A major ecological region containing a group of ecosystems characterized by its climate and characteristic organisms.

Biosphere- All the biomes on Earth. Thus, all the living areas of the planet.

Ecology- Population

POPULATION: Individuals of the same species living in a defined area at a particular time.

- ✓ All rice plants in a rice field =
Rice population
- ✓ Aphid population over bean field;
etc.

Ecology-Community

COMMUNITY: It is the natural assemblage of different populations in a particular habitat.

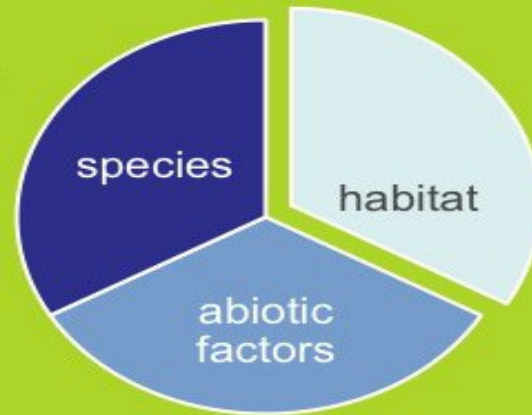
Examples: Pond community = (Water hyacinth + snails + fish).

Plant community = (Rice population + Jute population + Grass population etc.).

ECOSYSTEM

“Ecosystem”

- An ecosystem is the community of organisms in an area and the physical factors with which those organisms interact.



Abiotic factors: temperature, water, salinity, sunlight, or soil, temperature, precipitation, sunlight, and wind

BRANCHES OF ECOLOGY

Based on Levels of Ecological Organization:

Organismal ecology: An individual organism's responses and adaptations. For example, how a fish responds to low dissolved oxygen.

Population ecology: Individuals of the same species in an area; size, density and growth. For example, how the number of tilapia in a pond changes.

Community ecology: Interactions among populations of different species. For example, feeding relationships among fish, insects and plankton.

BRANCHES OF ECOLOGY

Ecosystem ecology: Organisms and the physical environment; energy flow and nutrient cycling. For example, How nutrients circulate among water, sediment, algae and fish.

Landscape ecology: Spatial patterns and connections among ecosystems. For example, how ponds, fields and wetlands are connected across an area.

Global ecology: Ecological processes across the planet. For example, how global climate change affects aquatic ecosystems.

Within organismal ecology

Behavioral ecology: how behavior affects survival and reproduction. For example, how fish choose feeding locations or avoid predators.

Physiological ecology: how body functions respond to environmental conditions. For example, changes in fish respiration when oxygen levels fall.

On the basis of traditional scope

Autecology: A particular species in relation to its environment.

- ✓ Answers to what conditions allow this species to survive, grow and reproduce?
- ✓ For example, environmental requirements of tilapia, including oxygen, temperature and food.

Synecology: A community in relation to its environment.

- ✓ Answers to how do species interact, and how does the environment shape the community?
- ✓ For example, relationships among fish, algae, plankton and insects in the pond.

On the basis of habitat:

Terrestrial ecology: land environments, such as forests, grasslands and deserts.

Aquatic ecology: water environments

- ✓ freshwater (ponds, rivers and lakes) ecology
- ✓ marine (seas and oceans) ecology,
- ✓ estuarine (where river water mixes with seawater) ecology.

Other branches of Ecology

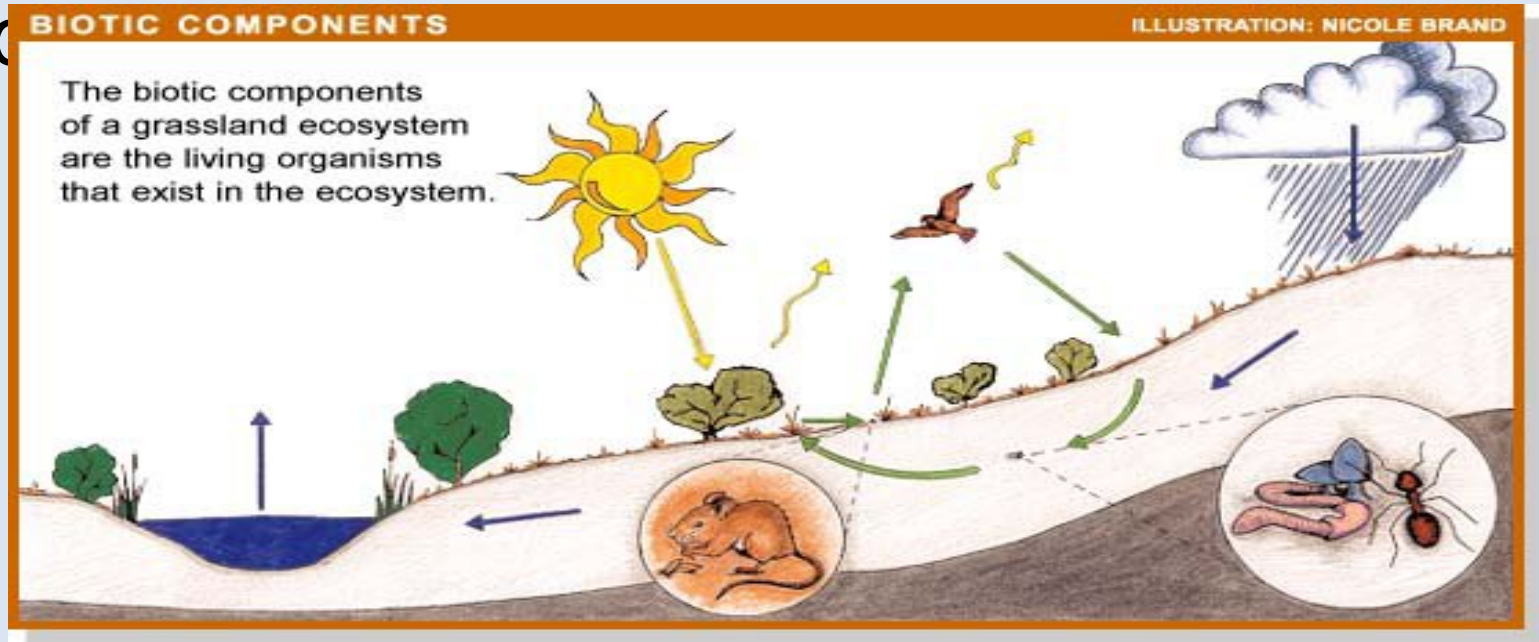
Systematic/Taxonomic position

- ✓ **Plant Ecology**, e.g. Crop, Forest etc.
- ✓ **Animal Ecology**, e.g. Mammalian Ecology, Insect Ecology, Fish Ecology, Algal Ecology, Microbial Ecology and so on.

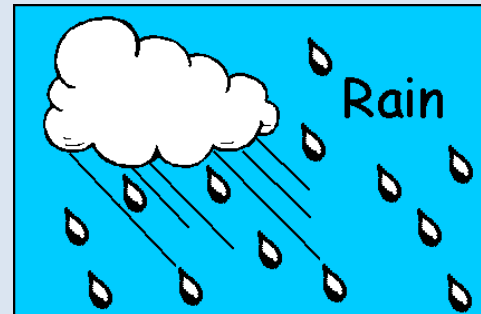
Application: Conservation ecology, Human ecology, Recreation ecology.

Method or approach: Mathematical and systems ecology. Models and analysis of ecological relationships and processes.

Biotic—living factors that influence an ecosystem



Abiotic—non-living factors that influence an ecosystem



ABIOTIC & BIOTIC FACTORS:

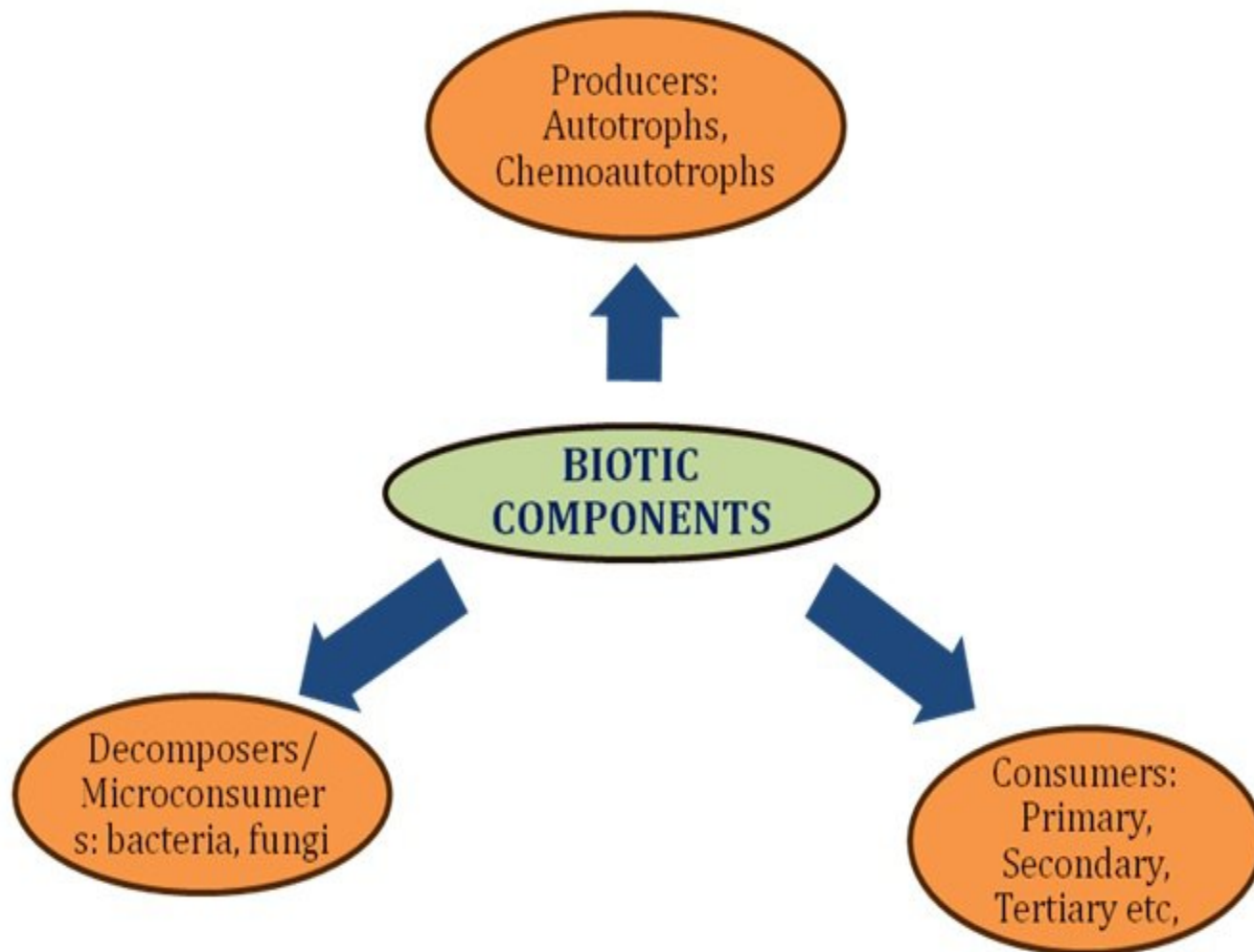
Abiotic and Biotic factors that influence ecosystems

Abiotic

- Air (O₂, CO₂, N₂, etc)
- Water
- Light
- Wind
- Soil
- pH
- Temperature
- Salinity
- Humidity
- Inorganic nutrients (N, P)
- Etc.

Biotic

- Other organisms, so:
- Competition
- Predation
- Symbiosis
 - Mutualism
 - Parasitism
- Disease agents



Producers

A. Sunlight is the main energy source for life on earth

B. Also called autotrophs

C. Use light or chemical energy to make food

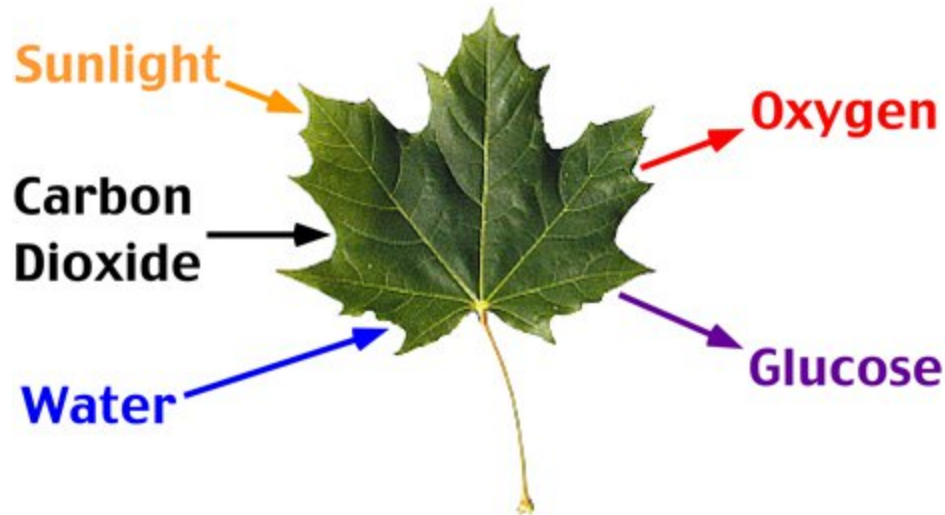
1. Plants
2. plant-like protists (algae)
3. Bacteria



D. Photosynthesis—use light energy to convert carbon dioxide and water into oxygen and carbohydrates

Light Energy
→

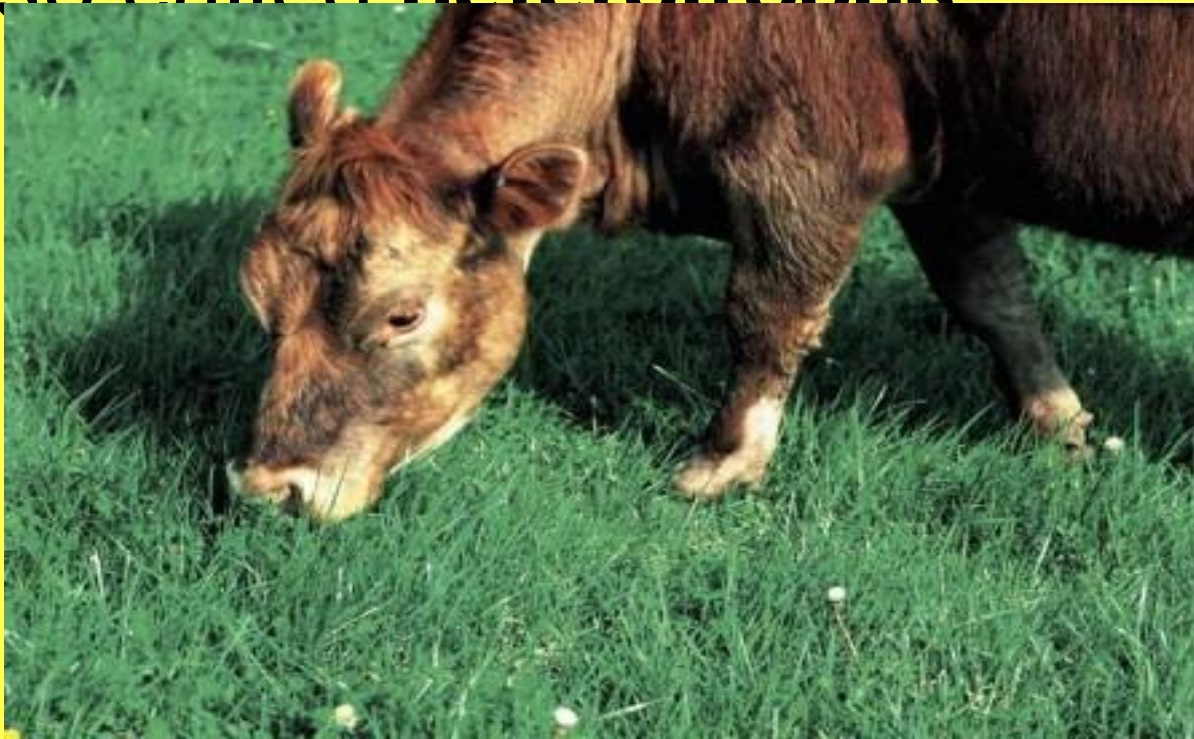
(Remember: $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow 6\text{O}_2 + \text{C}_6\text{H}_{12}\text{O}_6$)



E. Chemosynthesis—performed by bacteria, use chemical energy to produce carbohydrates

Consumers

- A. Organisms that rely on other organisms for their energy and food supply
- B. Also called **heterotrophs**



Herbivores — obtain energy
by
eating **only plants**



Carnivores — eat **only animals**

Omnivores—eat **both** plants and animals



Decomposers—breaks down dead organic matter

Feeding Interactions

- A. Energy flows through an ecosystem in **one direction** — **from the sun** or inorganic compounds **to autotrophs** (**producers**) and then **to heterotrophs** (**consumers**)

D. Trophic Levels—each step in a food chain or food web

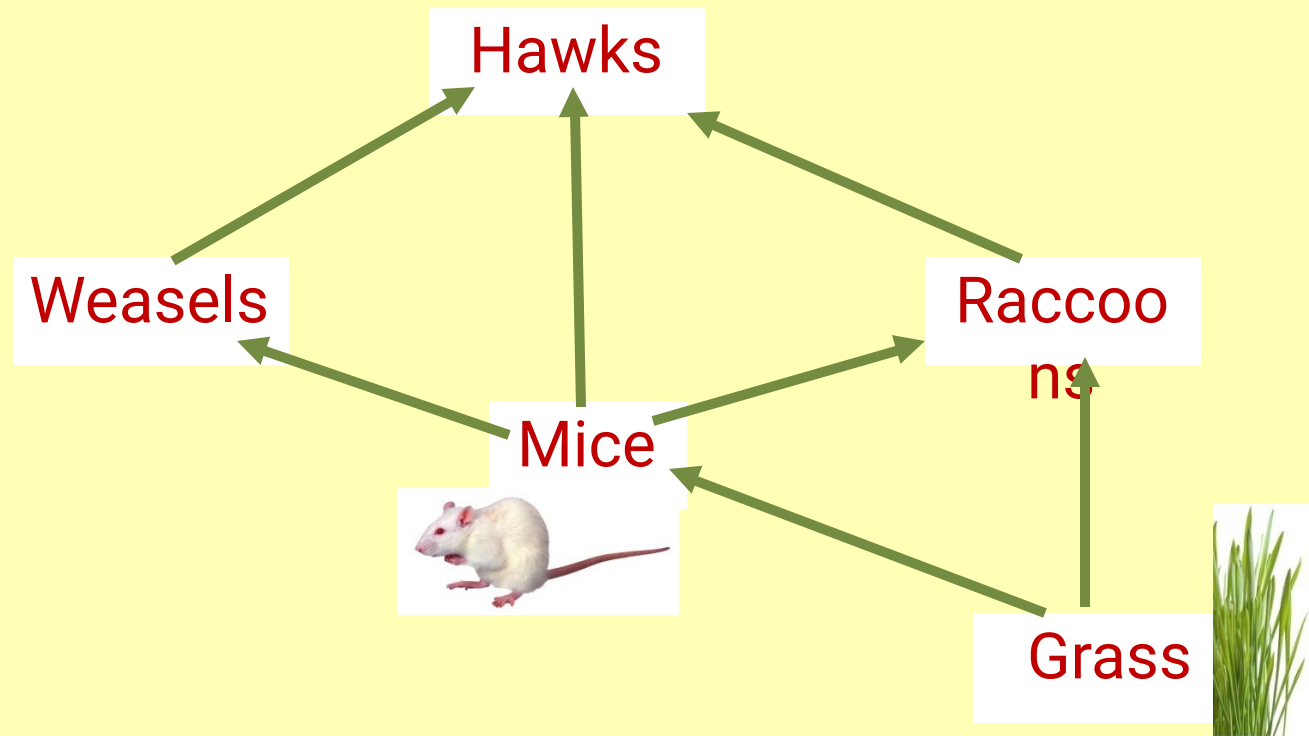
1. Level 1—**Producers** (autotrophs)

2. Level 2—**Primary** Consumers (**herbivores**)

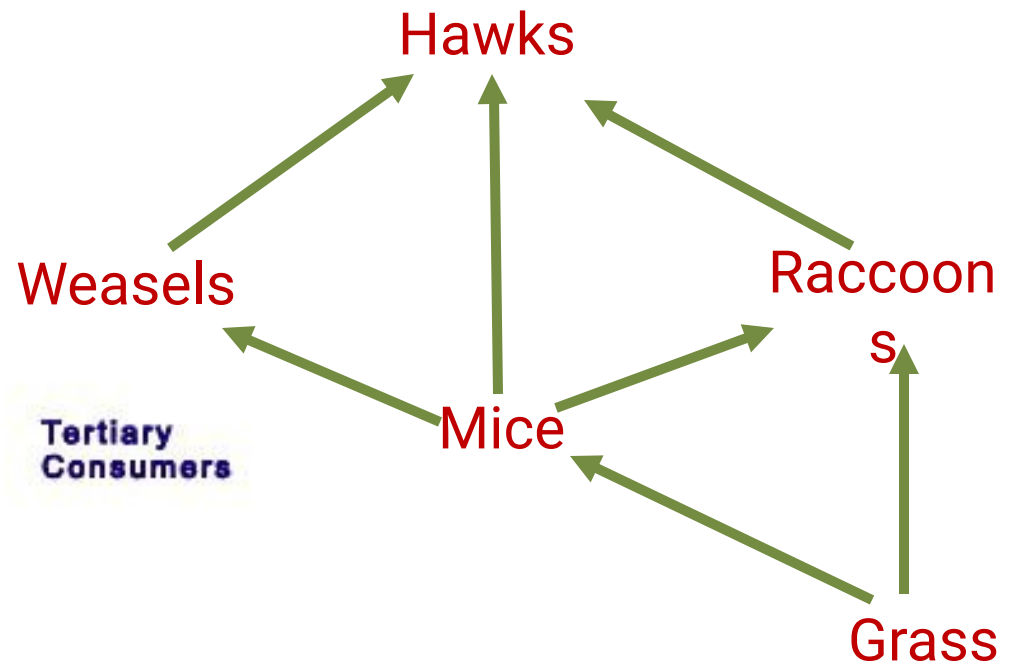
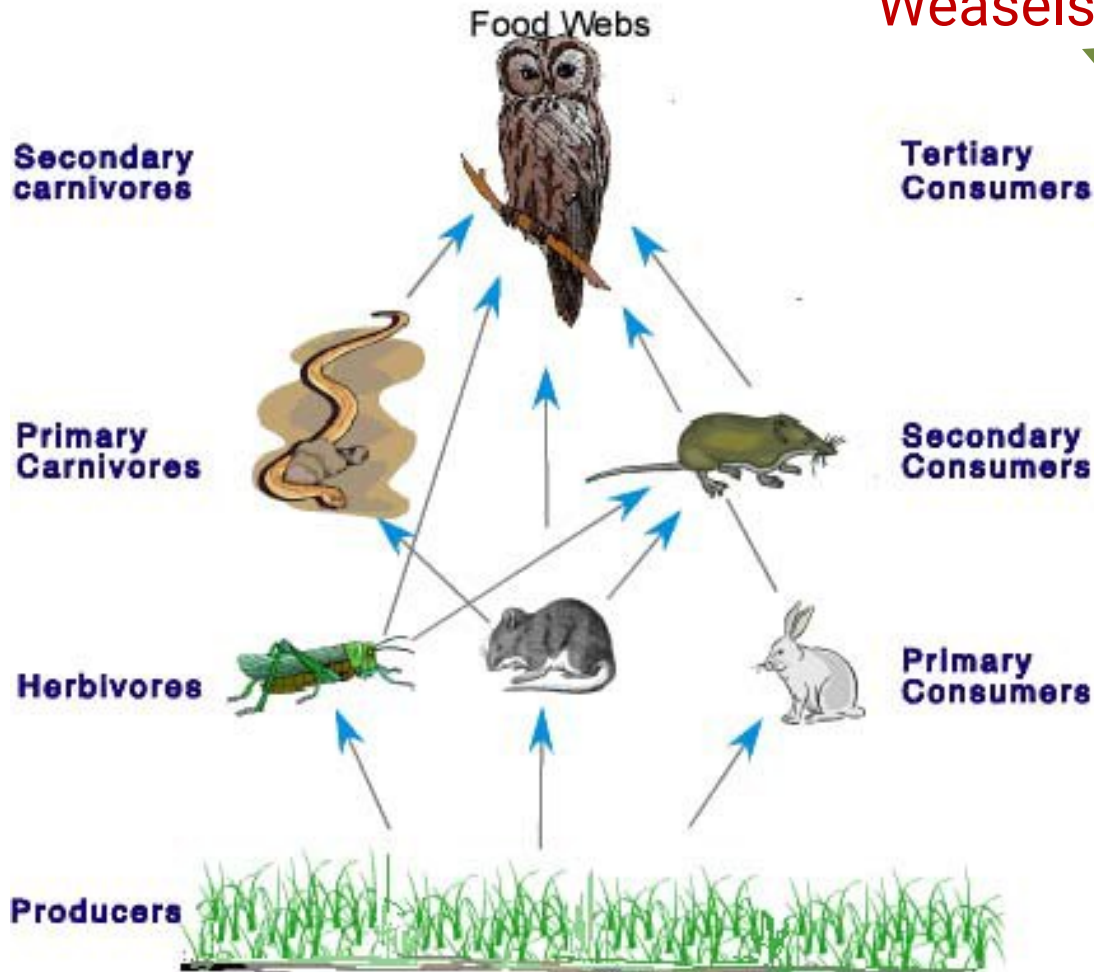
3. Level 3—**Secondary** Consumers
(**carnivores or omnivores**)

4. Level 4—**Tertiary** Consumers
(carnivore—usually **top carnivore**)

C. **Food Web**—**network** of food **chains** within an ecosystem



Food Webs

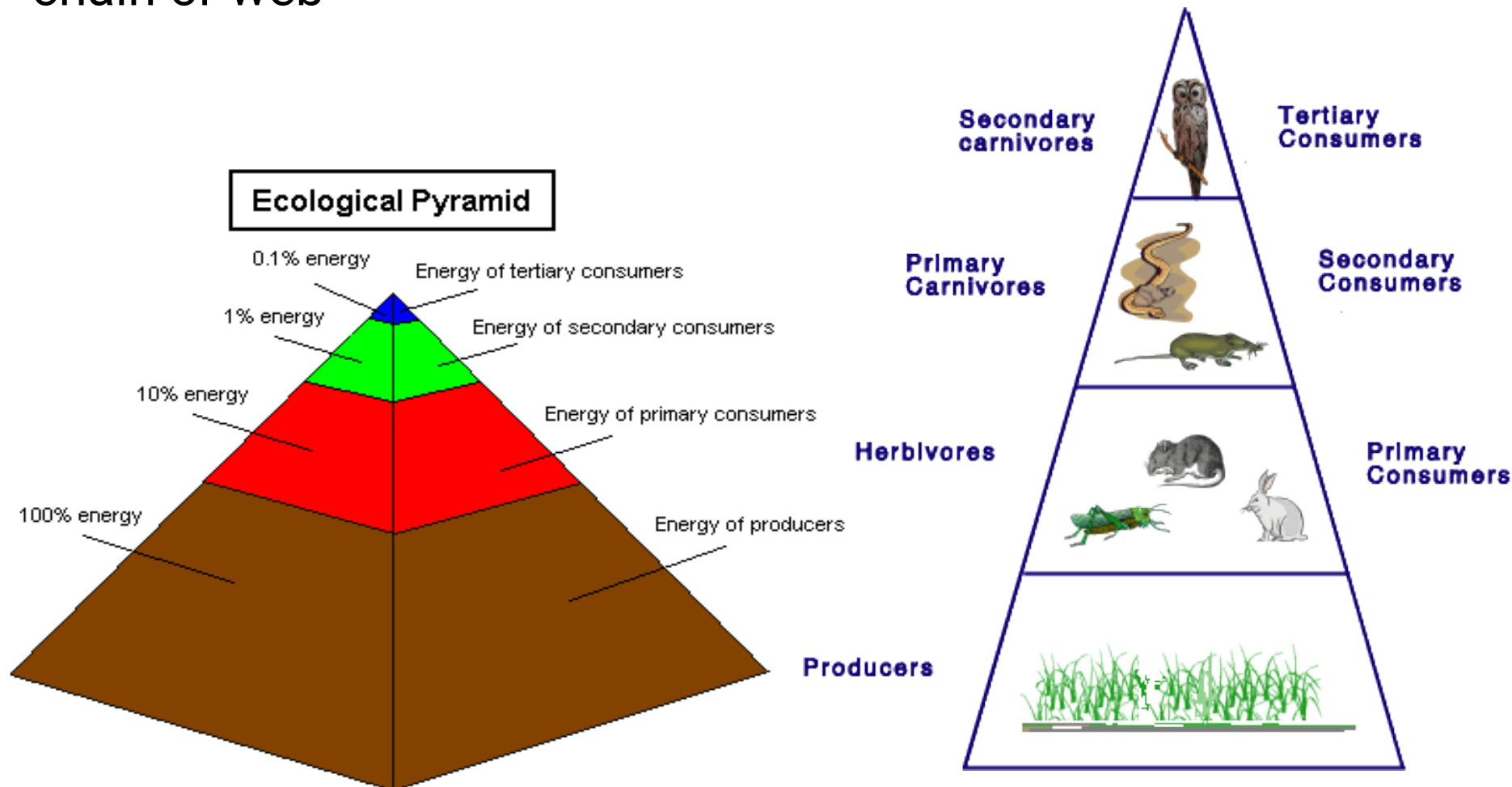


Energy flow and Nutrient Cycling

- Flow of energy:
- The energy used by the plants passes through various organisms and is ultimately lost as heat. **It is called the flow of energy.**
- Biogeochemical Cycling:
- Along with the energy, the chemical substances also pass through the organisms but become available for use again and again. Therefore, **the process is known as biogeochemical cycling.**
- The two processes bind the organisms (**eating** and **being eaten**) and their environment into an interacting complex unit—**the Ecosystem.**

IV. Ecological Pyramids

A. Diagram that shows the relative amount of **energy** or **organisms** contained within each **trophic level** of a food chain or web

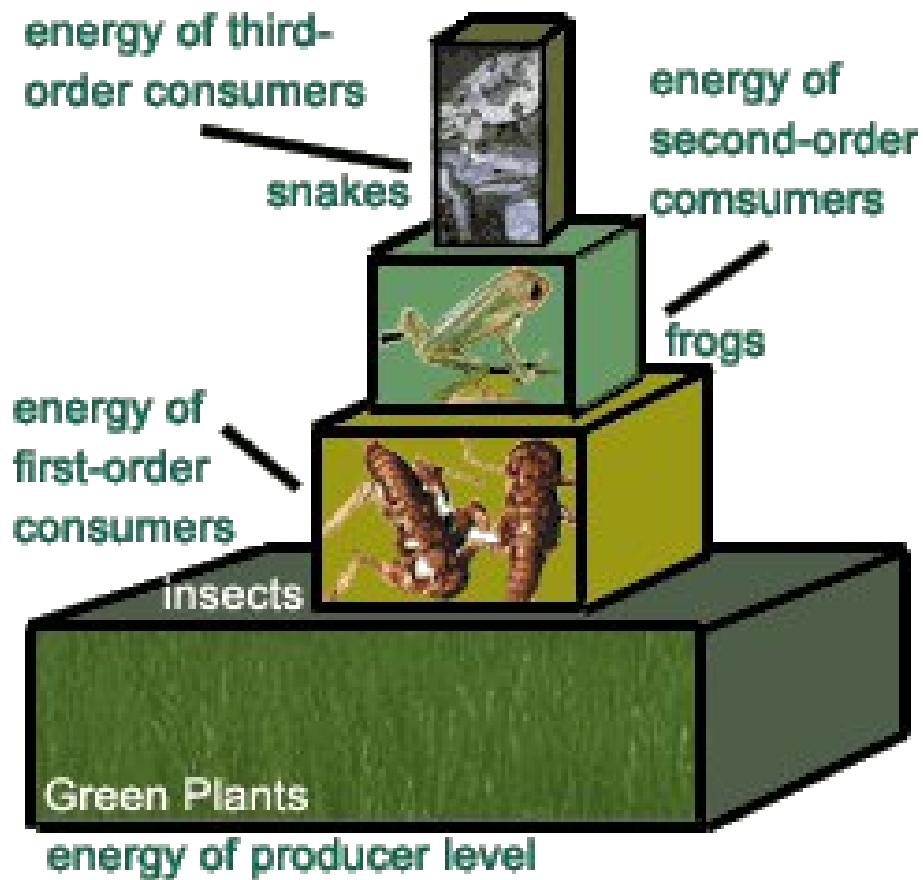


B. Energy Pyramid

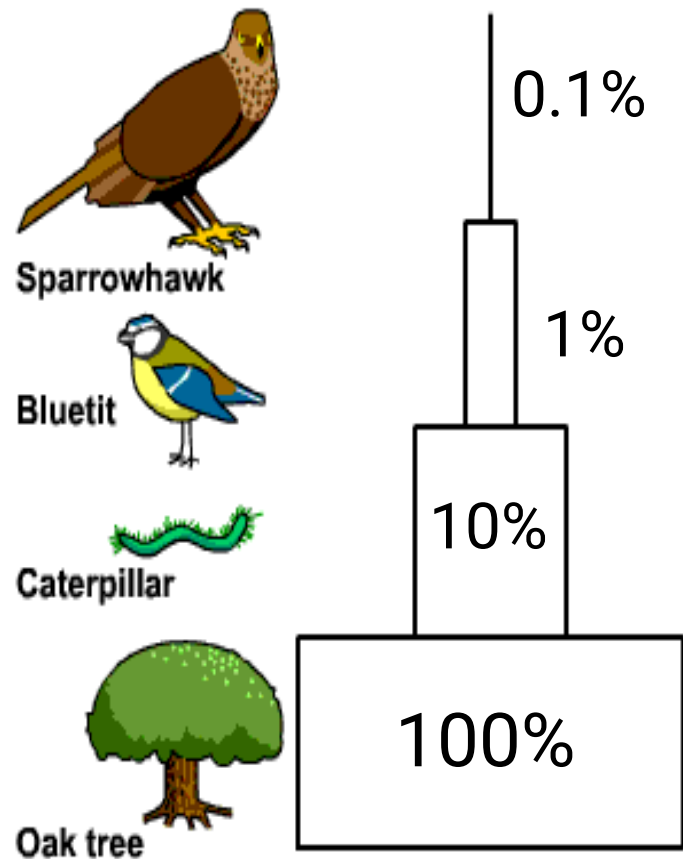
- Shows the energy available at each trophic level over a given period.
- Organisms use energy for life processes, including maintenance, growth and reproduction.
- Much of this energy is eventually released as heat through metabolism.
- Approximate 10% rule: About 10% of the energy at one trophic level transfers to the next higher level. The actual percentage varies.
- Energy decreases at higher trophic levels, so energy pyramids are always upright.

C. Biomass Pyramid

- Shows the total mass of living organic matter at each trophic level at a given time, usually expressed as dry mass per unit area (g/m^2).
- Biomass pyramids are usually upright in terrestrial ecosystems.
- They can be inverted in aquatic ecosystems, where producers reproduce rapidly despite having a small standing biomass.

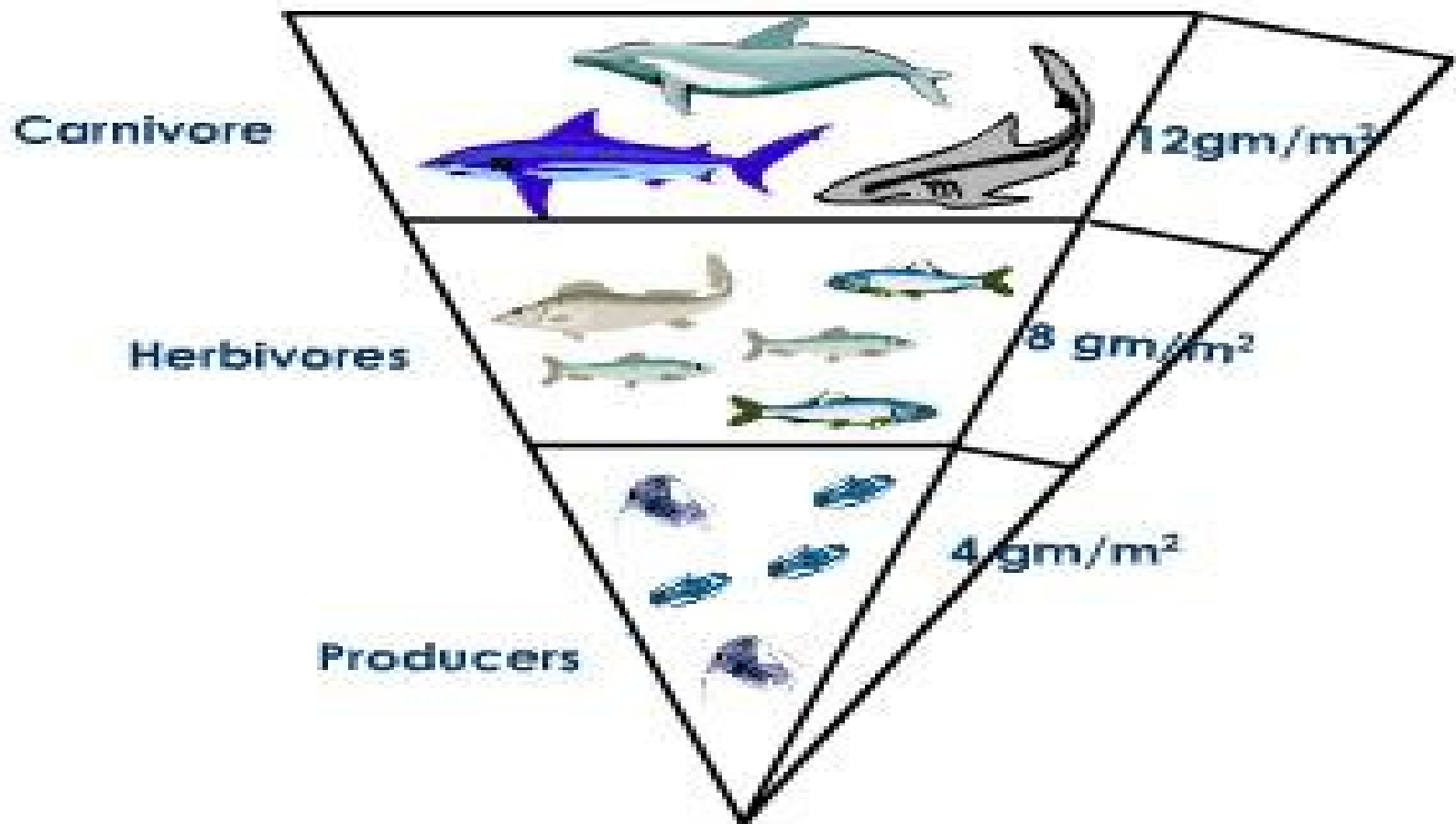


Energy
Pyramid



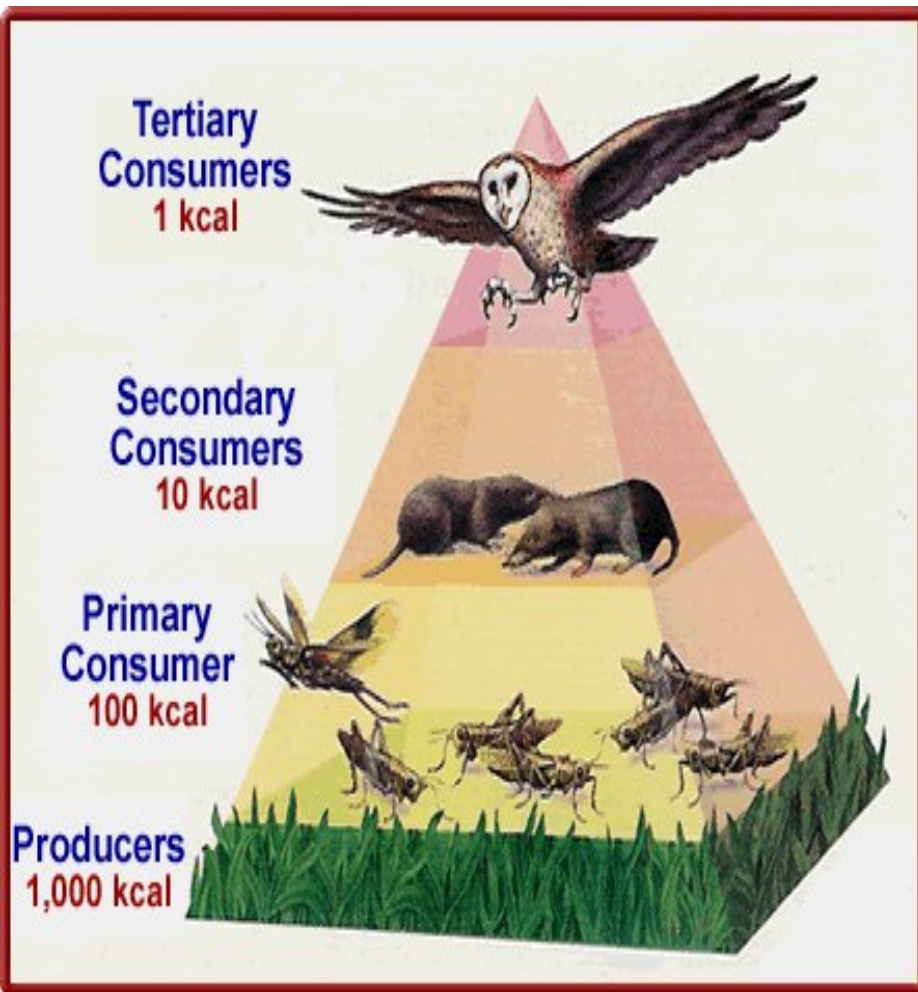
Number
Pyramid

Inverted Pyramid of Biomass

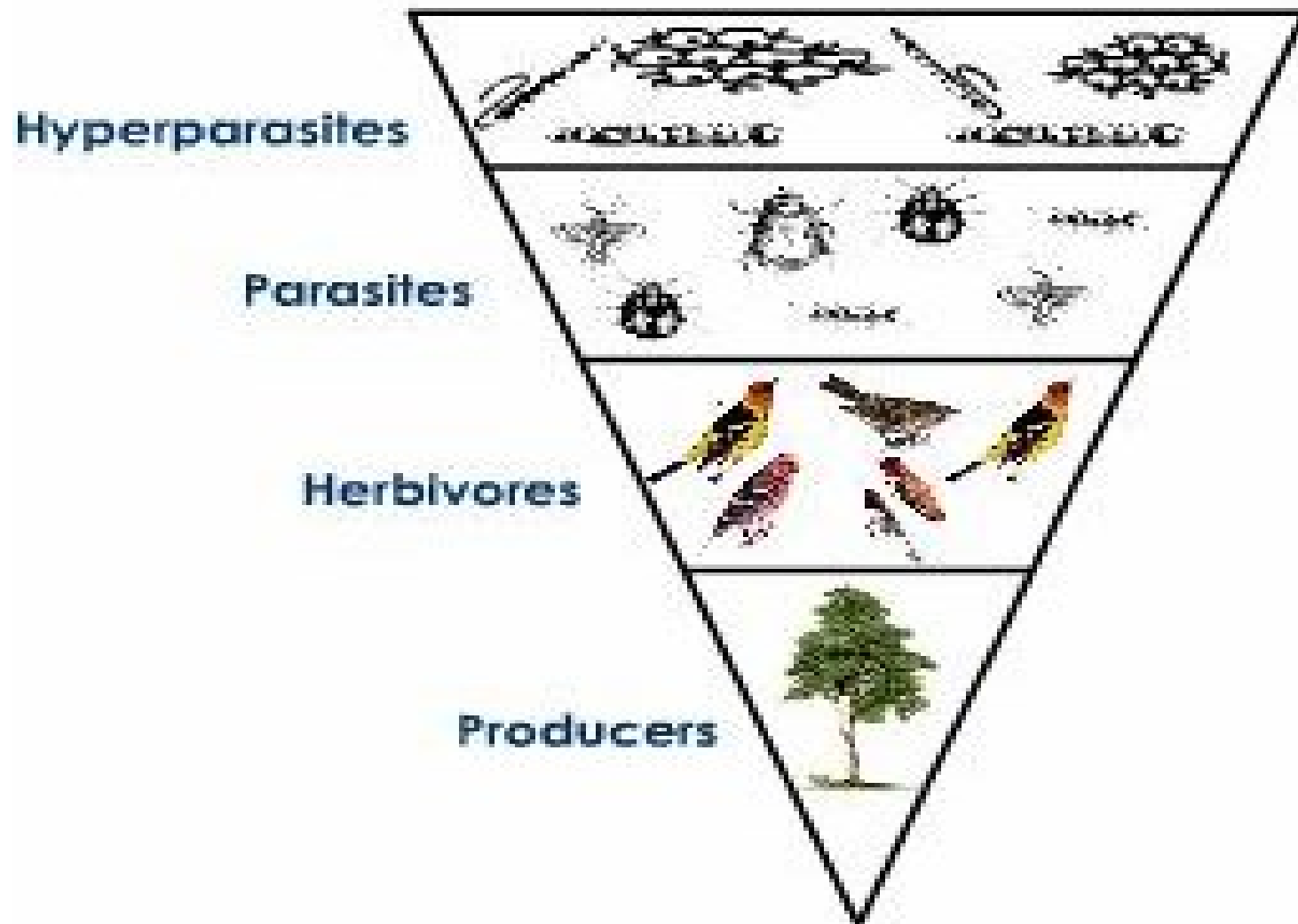


Inverted Pyramid in an Aquatic Ecosystem

Energy and Biomass Pyramid (together)



Represents amount of energy available at each level as well as amount of living tissue—both decrease with each increasing trophic level



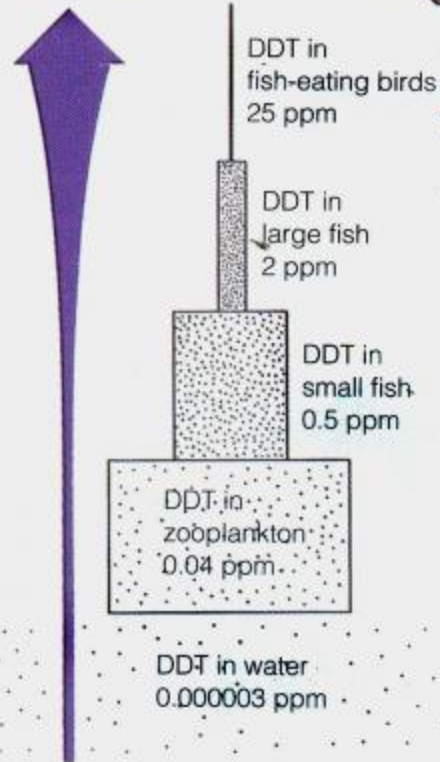
Inverted pyramid of number

Feeding relationships transfer both energy and pollutants

- ✓ Some pollutants enter organisms through contaminated water or food. When organisms retain these substances, they can build up and pass to predators.
- ✓ **Bioaccumulation:** A pollutant builds up within an organism over time because uptake exceeds elimination.
- ✓ A fish gradually accumulates methylmercury in its tissues.
- ✓ **Biomagnification:** Pollutant concentration increases across successive trophic levels in a food chain.
- ✓ A predatory fish has a higher methylmercury concentration than its prey.

BIOMAGNIFICATION

DDT concentration:
increase of
10 million times



Heavy metals
Ex: methylmercury

Pesticides
Ex: DDT
(dichlorodiphenyltri
chloroethane)

Organochlorides:
Ex: PCBs

The Roles of Organisms in an Ecosystem

All of the organisms within an ecosystem have different roles. These roles are called **niches**.

An Organism's niche can have many dimensions and knowing the niches of an organism can help to explain why they act and interact the way they do.

To determine an organism's niche, you need to identify how an organism uses resources, responds to conditions and interacts with other organisms in the same ecosystem.

For example, what the fish eats, where it feeds, its environmental requirements and its relationships with predators and competitors in pond.

B. Niche—the ecological niche involves both the **place** where an organism lives and the **roles** that an organism has in its **habitat**.

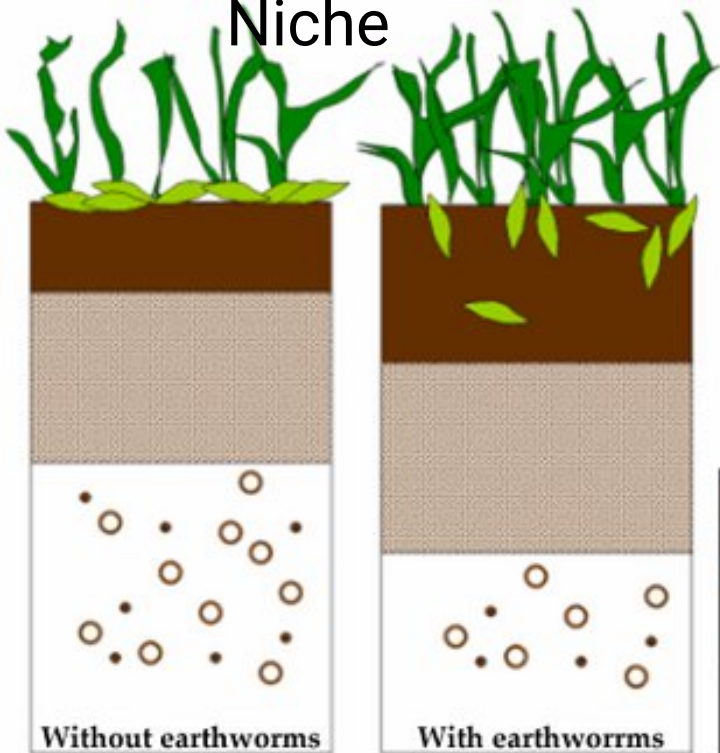
Example: The ecological niche of a **sunflower** growing in the backyard includes absorbing light, water and nutrients (for photosynthesis), providing shelter and food for other organisms (e.g. bees, ants, etc.), and giving off oxygen into the atmosphere.



The ecological niche of an organism depends not only on where it lives but also on what it does. By analogy, it may be said that the habitat is the organism's "address", and the niche is its "profession", biologically speaking.

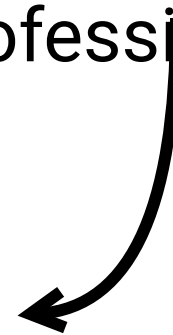
"Address"—Soil, Ground, etc.

Worm's
Niche



Enhanced plant yield
Less surface litter
More topsoil
More organic carbon, nitrogen and polysaccharides
Enhanced porosity, aeration and drainage

"Profession"— Mix-up soil



Ecological Interactions between organisms

A. Competition—Competition occurs when organisms use the same **limited resources**, such as food, water, light, nutrients, space or mates, reducing their availability to others.



Intraspecific competition

- ✓ Competition between individuals of the same species.
- ✓ Two rice plants competing for light, water and nutrients.

Interspecific competition

- ✓ Competition between individuals of different species.
- ✓ Rice plants and weeds competing for light, water and nutrients.



Monkeys compete with each other and other animals for food.

Rams compete with each other for mates.





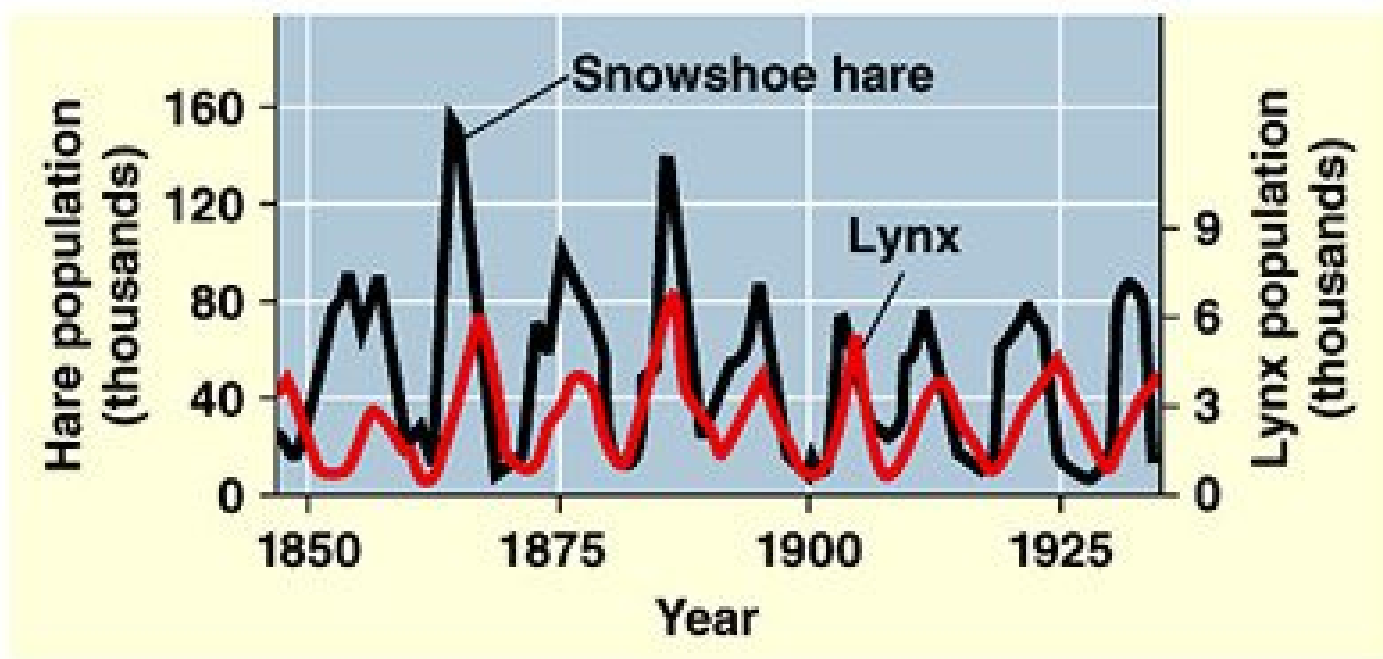
Until Americans introduced gray squirrels into parts of England in the early 20th century, red squirrels had been the only species of squirrel in the country. The gray squirrels were larger and bred faster and successfully competed for resources. Within a couple years of overlap in an area, the red squirrels disappeared.

B. **Predation**—one organism captures and feeds on another organism

1. **Predator**—one that does the killing

2. **Prey**—one that is the food





C. **Symbiosis**—any relationship in which two species live closely together

1. **Mutualism**—both species benefit (WIN-WIN)

a. Ex: **insects and flowers**

Can you think of any other examples that we've talked about in class?



2. **Commensalism**—one member of the association benefits and the other is neither helped nor harmed.

Example: **barnacles on a whale**





The Remora fish attaches to the shark and gets a free ride.

Commensalism

Birds build nests in trees.



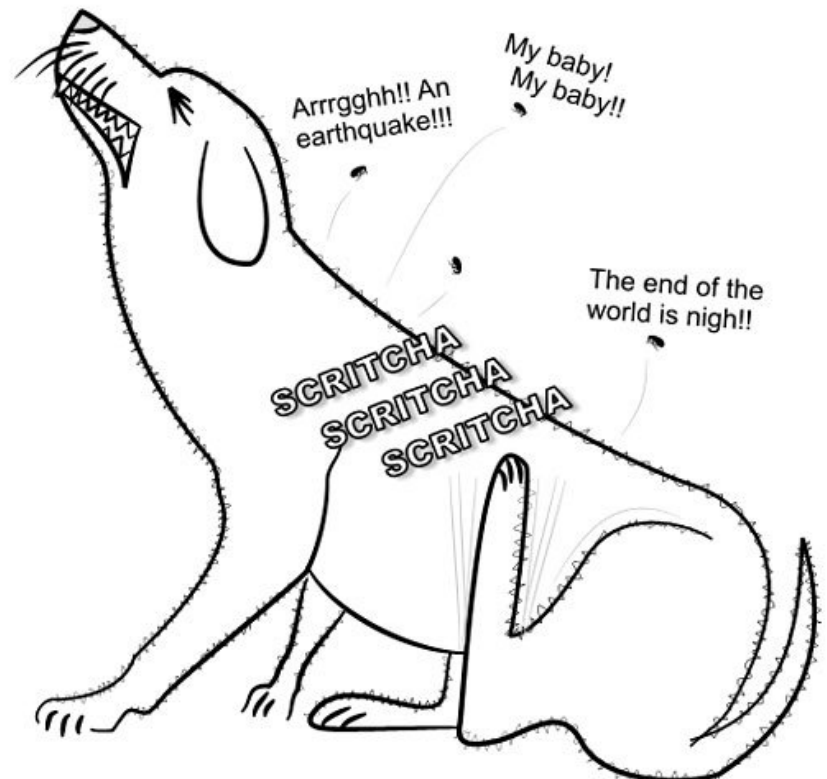
3. **Parasitism**—one organisms lives on or inside another organism (host) and harms it.

The parasite obtains all or part of its nutritional needs from the host. (WIN-LOSE)

Ex



dog





Wasp eggs on back
of caterpillar
(Parasitoids)

Parasitism



Sea lampreys feed
on fluids of other
fish.



Mosquito biting a
human.

4. Amensalism one species is harmed, while the other is essentially unaffected.

- ✓ For example, allelopathy in plants. Some plants release chemicals into the soil that inhibit the germination or growth of nearby plants. The neighbouring plant is harmed, while the chemical-producing plant is not directly affected by that interaction.
- ✓ For example, black walnut (*Juglans nigra*) releases juglone, which can suppress the growth of some nearby plant species.
- ✓ Another example is antibiosis: a microorganism produces a substance that inhibits another microorganism. For instance, *Penicillium* can produce antibiotics that inhibit bacteria.

Predation, Mutualism, Commensalism or Parasitism??

